

COHORT DAY 1 TASK

Small Language Model (SLM)

Introduction to SLM

A Small Language Model (SLM) is an artificial intelligence model designed to understand and generate human language while using fewer computational resources. SLMs are smaller in size compared to Large Language Models and are optimized for speed, efficiency, and deployment on devices with limited hardware. These models are trained on smaller datasets and require less memory for processing tasks. SLMs are commonly used in mobile applications, chatbots, virtual assistants, and embedded systems. They provide quick responses while consuming less power and storage. Organizations prefer SLMs when low latency and cost-effective deployment are important. Despite having fewer parameters, SLMs can perform language understanding, text generation, summarization, and question-answering tasks effectively. They are suitable for domain-specific applications where general knowledge is not the primary requirement. The development of SLMs helps make AI technology more accessible and affordable. They also offer better privacy because they can run locally on user devices. As AI technology advances, SLMs continue to improve in performance and efficiency.

Large Language Model (LLM)

Introduction to LLM

A Large Language Model (LLM) is an advanced artificial intelligence system trained on massive amounts of text data to understand and generate human-like language. LLMs contain billions of parameters that help them learn complex language patterns, reasoning abilities, and contextual understanding. These models are capable of performing a wide range of tasks such as content generation, translation, coding assistance, summarization, and conversational AI. Popular examples include GPT-based models and other transformer-based architectures. LLMs require powerful hardware and significant computational resources for training and inference. Their large knowledge base enables them to answer questions from various domains. They can understand context, generate detailed explanations, and perform complex reasoning tasks. LLMs are widely used in education, healthcare, business automation, and software development. The ability to process vast amounts of information makes them highly versatile. However, they may require more storage, memory, and operational costs. Continuous improvements in LLM technology are driving innovation across industries and transforming human-computer interaction.

Retrieval-Augmented Generation (RAG) System

Introduction to RAG System

Retrieval-Augmented Generation (RAG) is an AI architecture that combines information retrieval with language generation to provide accurate and context-aware responses. Instead of relying only on the knowledge stored in a language model, a RAG system first retrieves relevant information from external documents, databases, or knowledge sources. The retrieved content is then provided to the language model as additional context for generating responses. This approach improves accuracy and reduces hallucinations in AI-generated outputs. RAG systems are widely used in question-answering platforms, enterprise knowledge bases, customer support systems, and research applications. The retrieval component searches for the most relevant information based on user queries. The generation component uses the retrieved data to create meaningful and informative responses. RAG enables AI systems to access up-to-date information without retraining the model. It improves reliability and ensures responses are grounded in actual data sources. The architecture consists of data storage, retrieval mechanisms, embedding models, and language generation models. As a result, RAG systems provide efficient, scalable, and trustworthy AI solutions for modern applications.

How SLM and LLM Work in a RAG System

In a RAG system, both SLMs and LLMs can act as the generation component. The process begins when a user submits a query. The retrieval module searches a database or document repository and finds relevant information. This information is converted into contextual input and passed to the language model. If an SLM is used, the system provides faster responses with lower computational requirements, making it suitable for lightweight applications and edge devices. If an LLM is used, the system generates more detailed and context-rich responses because of its larger knowledge and reasoning capabilities. The retrieved documents help both models produce accurate answers based on real data rather than relying only on their trained knowledge. Therefore, RAG enhances the performance of both SLMs and LLMs by supplying relevant external information and reducing incorrect or outdated responses.

Difference Between SLM and LLM

SLM (Small Language Model)

1. SLM contains fewer parameters and has a smaller model size.
2. It requires less memory and computational resources.
3. It provides faster responses with lower latency.
4. It is mainly used in mobile devices, chatbots, and edge applications.

LLM (Large Language Model)

1. LLM contains billions of parameters and has a larger model size.
2. It requires high memory and powerful hardware resources.
3. It can perform complex reasoning and generate detailed responses.
4. It is mainly used in advanced AI systems, research, and enterprise applications.